KidsPortal

Maths Curriculum & Syllabus Guide

Grades 1 – 10 · A reference for parents

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About this guide

This document lists every topic your child covers in KidsPortal's Maths subject, grade by grade, so you can see exactly what is being taught and when. Maths is benchmarked primarily to Cambridge Primary Maths (number, geometry and measure, statistics, thinking mathematically), and also checked against England's National Curriculum for Mathematics — with the roughly equivalent US grade band noted for families comparing against Common Core Math standards. These alignment notes describe the learning stage and skill focus each framework shares with that grade, not a word-for-word copy of any curriculum's official programme of study.

Every topic below is a real, live lesson on KidsPortal — this snapshot lists what your child actually covers, grade by grade. The curriculum is organised into four learning bands (Seedlings, Explorers, Navigators, Pathfinders) that group related skills together regardless of age, so a lesson always matches what your child is ready for. For the most current version, use the interactive curriculum browser on the Curriculum page — this PDF is a point-in-time copy.

Grade 1

B1 · Seedlings band

Place value to 100, the four operations in their simplest form, fractions, measurement, money, time and shape.

1. Place Value to 100
2. Addition & Subtraction
3. Multiplication & Division
4. Fractions
5. Measurement
6. Money
7. Time
8. Shape
9. Position & Direction
10. Statistics
11. Counting & Number Bonds to 20
12. Doubling, Halving & Sharing
13. Comparing & Ordering Numbers
14. Odd & Even Numbers
15. Simple Addition & Subtraction Word Problems
16. Patterns & Sequences
17. Comparing Lengths & Heights
18. Counting Coins & Simple Money
19. Comparing Capacity (More/Less Full)
20. 2D Shape Sorting
21. Telling Time to the Hour & Half Hour
22. Grouping & Counting in Tens

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 1 (Year 1) National Curriculum for Mathematics — counting, number bonds and simple addition/subtraction — broadly equivalent to US Kindergarten early-numeracy expectations. This is our Seedlings band, benchmarked primarily to Cambridge Primary Maths (number, geometry and measure, statistics, thinking mathematically).

“count to 100 and show tens and ones”

“add and subtract in my head”

“tell the time and count coins”

Grade 2

B1 · Seedlings band

Place value to 1,000, formal addition and subtraction, the 3, 4 and 8 times tables, fractions of amounts, and simple data handling.

1. Place Value to 1,000
2. Formal Addition & Subtraction
3. Times Tables: 3, 4 & 8
4. Tenths & Equivalent Fractions
5. Perimeter
6. Money: Problem Solving
7. Time
8. Angles & Shape Properties
9. Bar Charts & Tables
10. Reasoning & Problem Solving
11. Multiplication & Division Facts: 2, 5 & 10
12. Fractions of Shapes & Amounts
13. Comparing Mass & Capacity
14. 3 & 4 Times Tables
15. 2D Shape Properties: Sides & Vertices
16. Telling Time to 5 Minutes
17. Division as Sharing Equally
18. Introducing Halves & Quarters
19. Money: Adding Coins and Notes
20. Measuring Length in cm and m
21. Weighing Objects in Grams and Kilograms
22. Simple Tally Charts and Data

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 1 (Year 2) National Curriculum for Mathematics, including the statutory 2, 5 and 10 times tables — broadly equivalent to US Grade 1 Common Core Math standards. This is our Seedlings band, benchmarked primarily to Cambridge Primary Maths (number, geometry and measure, statistics, thinking mathematically).

“count to 100 and show tens and ones”

“add and subtract in my head”

“tell the time and count coins”

Grade 3

B2 · Explorers band

Place value to 10,000, negative numbers, times tables to 12×12, decimals, area and perimeter, and multi-step word problems.

1. Place Value to 10,000
2. Negative Numbers
3. Times Tables to 12×12
4. Decimals
5. Area & Perimeter
6. Money & Decimals
7. Converting Units of Time
8. Symmetry & Coordinates
9. Discrete & Continuous Data
10. Multi-Step Word Problems
11. Formal Column Addition & Subtraction
12. Adding & Subtracting Fractions
13. Telling Time: Digital & 24-Hour Clock
14. Multiplying 2-Digit by 1-Digit Numbers
15. Perimeter of Simple Shapes
16. Bar Charts & Pictograms
17. Equivalent Fractions
18. Measuring in Litres & Millilitres
19. Telling Time: the 24-Hour Clock
20. Finding Area by Counting Squares
21. Multi-Step Word Problems
22. Introducing Roman Numerals

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 2 (Year 3) National Curriculum for Mathematics and its full times-tables-to-12×12 expectation — broadly equivalent to US Grade 2 Common Core Math standards. This is our Explorers band, benchmarked primarily to Cambridge Primary Maths (number, geometry and measure, statistics, thinking mathematically).

“read and write big numbers and decimals”

“multiply and divide using written methods”

“explain how I solved a problem”

Grade 4

B2 · Explorers band

Place value to a million, formal written multiplication and division, fractions and decimals, perimeter/area/volume, and line graphs.

1. Place Value to 1,000,000
2. Roman Numerals
3. Formal Written Multiplication & Division
4. Adding & Subtracting Fractions
5. Decimals & Percentages
6. Perimeter, Area & Volume
7. Angles in Shapes
8. Position & Transformation
9. Line Graphs
10. Problem Solving & Reasoning
11. Multiplying & Dividing by 10, 100 & 1,000
12. Rounding & Estimating
13. Properties of Triangles & Quadrilaterals
14. Introducing Decimals: Tenths & Hundredths
15. Area of Rectangles
16. Negative Numbers on a Number Line
17. Adding & Subtracting Fractions with the Same Denominator
18. Line Symmetry
19. Rounding Decimals
20. Decimals in Money and Measurement
21. Plotting Coordinates (First Quadrant)
22. Translating Shapes

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 2 (Year 4) National Curriculum for Mathematics, where formal written methods for all four operations are introduced — broadly equivalent to US Grade 3 Common Core Math standards. This is our Explorers band, benchmarked primarily to Cambridge Primary Maths (number, geometry and measure, statistics, thinking mathematically).

“read and write big numbers and decimals”

“multiply and divide using written methods”

“explain how I solved a problem”

Grade 5

B2 · Explorers band

Place value to 10 million, the four operations with larger numbers, ratio and proportion, introductory algebra, and compound shapes.

1. Place Value to 10,000,000
2. The Four Operations
3. Fractions, Decimals & Percentages
4. Ratio & Proportion
5. Introduction to Algebra
6. Metric & Imperial Conversion
7. Compound Shapes
8. Classifying Shapes
9. Pie Charts & the Mean
10. Reasoning Papers
11. Prime Numbers, Factors & Multiples
12. Multiplying & Dividing Fractions
13. Volume of Cubes & Cuboids
14. Introducing Ratio
15. Angles in a Straight Line & Around a Point
16. Mean, Median, Mode & Range
17. Percentages of Amounts
18. Plotting Coordinates in All Four Quadrants
19. Converting Between Fractions, Decimals & Percentages
20. Volume of Cubes and Cuboids
21. Finding the Mean (Average)
22. Simple Formulae and Substitution

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 2 (Year 5) National Curriculum for Mathematics, including the first introduction to algebraic notation — broadly equivalent to US Grade 4 Common Core Math standards. This is our Explorers band, benchmarked primarily to Cambridge Primary Maths (number, geometry and measure, statistics, thinking mathematically).

“read and write big numbers and decimals”

“multiply and divide using written methods”

“explain how I solved a problem”

Grade 6

B3 · Navigators band

Algebraic expressions, ratio and proportion, angle and shape properties, probability basics, and solving linear equations.

1. Number
2. Algebraic Expressions
3. Fractions, Decimals & Percentages
4. Ratio & Proportion
5. Angles & Shape Properties
6. Averages & Range
7. Probability Basics
8. Sequences
9. Solving Linear Equations
10. Area, Perimeter & Volume
11. Negative Numbers & Order of Operations (BIDMAS)
12. Percentages: Increase & Decrease
13. Constructions & Loci
14. Simplifying Algebraic Expressions
15. Circumference & Area of a Circle
16. Pie Charts
17. Solving One-Step & Two-Step Equations
18. Volume of Cuboids
19. Nets of 3D Shapes
20. Ratio and Proportion Problems
21. Combined Probability Events
22. Negative Number Operations

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 2 (Year 6) National Curriculum for Mathematics, the final year of primary maths before secondary school — broadly equivalent to US Grade 5 Common Core Math standards. This is our Navigators band, benchmarked primarily to Cambridge Lower Secondary Maths.

“work with negative numbers and powers”

“solve an equation for x”

“work out the chance of something happening”

Grade 7

B3 · Navigators band

Indices and standard form, Pythagoras' theorem, scatter graphs and correlation, transformations, and equations and inequalities.

1. Indices & Standard Form
2. Expanding & Factorising
3. Ratio, Proportion & Rates of Change
4. Pythagoras' Theorem
5. Scatter Graphs & Correlation
6. Combined Probability Events
7. Linear Graphs & Coordinates
8. Transformations
9. Equations & Inequalities
10. Compound Measures
11. Direct & Inverse Proportion
12. Congruence & Similarity
13. Solving Quadratic Equations by Factorising
14. Plotting & Interpreting Linear Graphs (y = mx + c)
15. Probability Basics
16. Standard Form
17. Finding the nth Term of Linear Sequences
18. Solving Linear Inequalities
19. Compound Measures: Speed, Density & Pressure
20. Factorising Simple Expressions
21. Scatter Graphs and Correlation
22. Congruence and Similarity

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 3 (Year 7) National Curriculum for Mathematics, where secondary pupils meet standard form and formal proof-style reasoning — broadly equivalent to US Grade 6 Common Core Math standards. This is our Navigators band, benchmarked primarily to Cambridge Lower Secondary Maths.

“work with negative numbers and powers”

“solve an equation for x”

“work out the chance of something happening”

Grade 8

B3 · Navigators band

Surds, quadratic expressions, simultaneous equations, introductory trigonometry, circle theorems and vectors — GCSE foundations.

1. Surds & Standard Form
2. Quadratic Expressions
3. Simultaneous Equations
4. Pythagoras & Introduction to Trigonometry
5. Sampling & Data Analysis
6. Probability Trees & Venn Diagrams
7. Quadratic & Real-Life Graphs
8. Circle Theorems
9. Vectors
10. GCSE Foundations Review
11. Trigonometric Graphs: Sine, Cosine & Tangent
12. Algebraic Fractions
13. Histograms & Frequency Density
14. Solving Simultaneous Linear Equations
15. Introduction to Vectors
16. Pythagoras' Theorem in 3D
17. Expanding Two Brackets
18. Introduction to Trigonometry (SOHCAHTOA)
19. Direct & Inverse Proportion
20. Solving Simultaneous Equations Graphically
21. Circle Area and Circumference Problems
22. Standard Form Calculations

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 3 (Year 8) National Curriculum for Mathematics; many schools begin GCSE-specification topics like circle theorems and vectors here, as this unit does — broadly equivalent to US Grade 7 Common Core Math standards. This is our Navigators band, benchmarked primarily to Cambridge Lower Secondary Maths.

“work with negative numbers and powers”

“solve an equation for x”

“work out the chance of something happening”

Grade 9

B4 · Pathfinders band

Standard form, quadratics, circle theorems, trigonometry beyond right angles, sampling and data, and functions and graphs.

1. Number: Standard Form, Surds & Indices
2. Algebra: Expanding, Factorising & Quadratics
3. Ratio, Proportion & Rates of Change
4. Geometry: Circle Theorems
5. Geometry: Trigonometry Beyond Right Angles
6. Statistics: Sampling & Data Representation
7. Probability: Combined Events & Tree Diagrams
8. Algebra: Simultaneous Equations
9. Vectors
10. Functions & Graphs
11. The Sine Rule & Cosine Rule
12. Completing the Square
13. Compound Interest & Growth/Decay
14. Function Notation
15. Transformations of Graphs
16. Surds
17. Solving Quadratic Equations by Factorising
18. Circle Theorems Introduction
19. Trigonometric Graphs
20. Simplifying Algebraic Fractions
21. Introduction to the Sine and Cosine Rules
22. Introduction to Histograms

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Still within Key Stage 3 (Year 9) nationally, though schools commonly start GCSE Maths content here to spread the specification over three years, as this unit does — broadly equivalent to US Grade 8 Common Core Math standards. This is our Pathfinders band, benchmarked primarily to Cambridge Lower Secondary Maths at IGCSE-adjacent standard.

“solve a quadratic equation”

“use trigonometry to solve triangles”

“analyse a data set and its spread”

Grade 10

B4 · Pathfinders band

Algebraic proof, 3D geometry, cumulative frequency and box plots, conditional probability, and an introduction to differentiation and integration.

1. Algebra: Algebraic Fractions & Proof
2. Number: Bounds & Error Intervals
3. Geometry: 3D Shapes, Volume & Surface Area
4. Geometry: Similarity & Congruence
5. Statistics: Cumulative Frequency & Box Plots
6. Probability: Conditional Probability
7. Algebra: Iterative Methods & Graph Transformations
8. Ratio: Direct & Inverse Proportion
9. Transformations of Functions
10. Exam Revision: Problem-Solving & Mixed Practice
11. Vectors: Geometric Proof
12. Advanced Circle Theorems
13. Simultaneous Equations: One Linear, One Quadratic
14. Introduction to Differentiation
15. Cumulative Frequency & Box Plots
16. Algebraic Proof
17. Differentiation of Polynomials (Further Practice)
18. Integration Basics
19. Vector Geometry Problems
20. Differentiation Applications: Finding Maximum and Minimum Points
21. Composite and Inverse Functions
22. Introduction to Iterative Methods

HOW THIS ALIGNS WITH REAL-WORLD CURRICULA

Matches the Key Stage 4 (GCSE Year 1) (Year 10) — GCSE Maths content including calculus foundations (differentiation and integration), usually reserved for higher-tier papers — broadly equivalent to US Grade 9 (Freshman) Common Core Math standards for that band. This is our Pathfinders band, benchmarked primarily to Cambridge Lower Secondary Maths at IGCSE-adjacent standard.

“solve a quadratic equation”

“use trigonometry to solve triangles”

“analyse a data set and its spread”